Needle-stick injuries among nursing students: a cross-sectional study

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Abstract

Background: Needle-stick injuries are common in training. Methods: This cross-sectional survey included 220 nursing students. Results: One in four reported an injury. Conclusion: Training should be strengthened.

Keywords: needle-stick injury; nursing students; occupational safety; cross-sectional study

1 Introduction

Needle-stick injuries expose health workers to blood-borne infections [1]. Students are at particular risk [2, 3].

2 Methods

This cross-sectional study was conducted between January 2025 and March 2025 and is reported according to STROBE. A total of 220 students ($n = 220$) took part; 176 (80.0%) were women. Inclusion criteria were second- to fourth-year students; exclusion criteria were students on leave. Injuries were recorded with a questionnaire, provided as Supplementary File 1. Statistical analysis used chi-square tests in R version 4.3; missing responses were excluded, and recall bias is discussed.

The study was approved by the Institutional Ethics Committee (approval no. IEC/2024/090, dated 5 December 2024) and followed the Declaration of Helsinki. Written informed consent was obtained from all participants.

3 Results

One in four students reported an injury (Table 1; Figure 1) [4].

Figure 1: Injuries by year of study.

Table 1: Participant characteristics.

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4 Discussion

Similar rates were reported elsewhere [4].

5 Acknowledgements

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7 Conflicts of interest

The authors declare no conflicts of interest.

8 Author contributions

PS: Conceptualization, Methodology, Writing – original draft. RV: Formal analysis, Writing – review & editing.

9 Data availability statement

The anonymised data are deposited in Zenodo (https://doi.org/10.5281/zenodo.2345678).

10 Declaration of generative AI use

No generative AI tools were used in preparing this manuscript.

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